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import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Generate base synthetic shape image
base_img = np.ones((400, 600), dtype=np.uint8) * 128
cv2.circle(base_img, (150, 200), 80, 220, -1)
cv2.rectangle(base_img, (320, 100), (520, 300), 50, -1)

# 2. Apply artificial illumination gradient ramp (bright left -> dark right)
x = np.linspace(1.8, 0.2, 600)
y = np.linspace(1.8, 0.2, 400)
xx, yy = np.meshgrid(x, y)
illumination_ramp = xx * 0.5
img_illuminated = np.clip(base_img.astype(np.float32) * illumination_ramp, 0, 255).astype(np.uint8)

# 3. Apply Edge Detection Strategies
# A. Direct Canny (Fixed thresholds)
edges_direct = cv2.Canny(img_illuminated, 50, 150)

# B. Global Histogram Equalization + Canny
img_he = cv2.equalizeHist(img_illuminated)
edges_he = cv2.Canny(img_he, 50, 150)

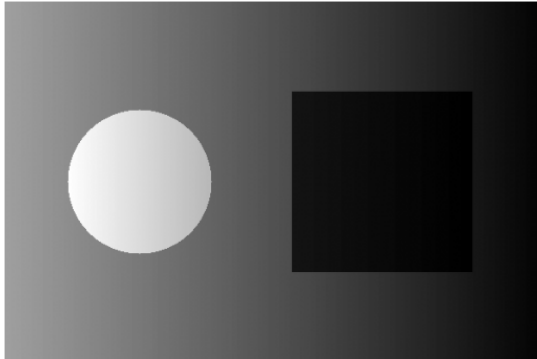
# C. CLAHE (Local Adaptive Equalization) + Canny
clahe = cv2.createCLAHE(clipLimit=3.0, tileGridSize=(8, 8))
img_clahe = clahe.apply(img_illuminated)
edges_clahe = cv2.Canny(img_clahe, 50, 150)

# 4. Display Visual Results
plt.figure(figsize=(16, 8))
plt.subplot(2, 3, 1); plt.imshow(img_illuminated, cmap='gray'); plt.title('1. Non-Uniform Illumination')
plt.subplot(2, 3, 2); plt.imshow(img_he, cmap='gray'); plt.title('2. Global HE Preprocessing'); plt
plt.subplot(2, 3, 3); plt.imshow(img_clahe, cmap='gray'); plt.title('3. CLAHE Local Normalization')
plt.subplot(2, 3, 4); plt.imshow(edges_direct, cmap='gray'); plt.title('4. Direct Canny (Shadow Free)')
plt.subplot(2, 3, 5); plt.imshow(edges_he, cmap='gray'); plt.title('5. Global HE + Canny (Noise Artifacts)')
plt.subplot(2, 3, 6); plt.imshow(edges_clahe, cmap='gray'); plt.title('6. CLAHE + Canny (Robust Canny)')

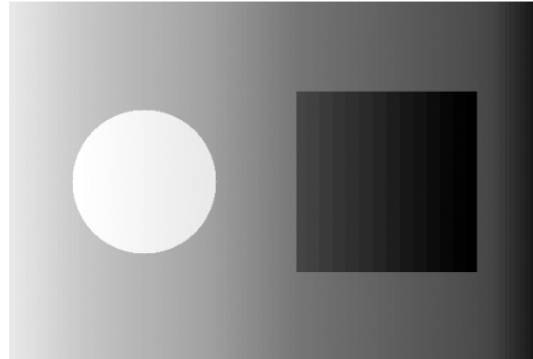
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plt.subplot(2, 3, 6), plt.imshow(edges_clahe, cmap= gray ), plt.title( 6. CLAHE + Canny (Robust Co  
plt.tight_layout(); plt.show()
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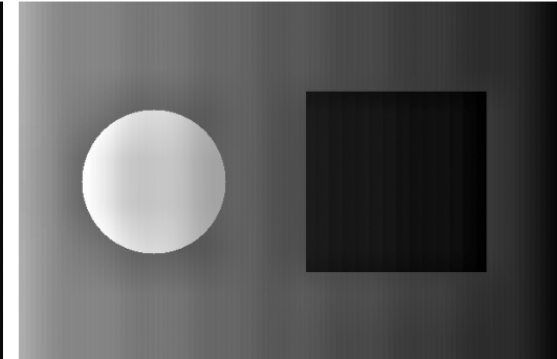
1. Non-Uniform Illumination Input



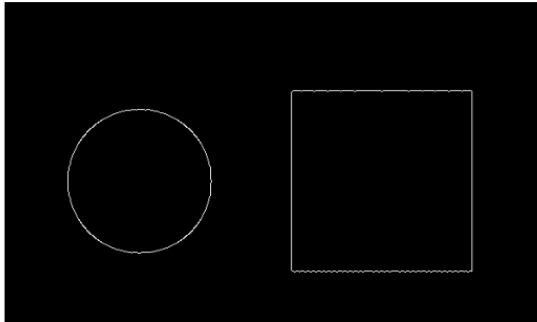
2. Global HE Preprocessing



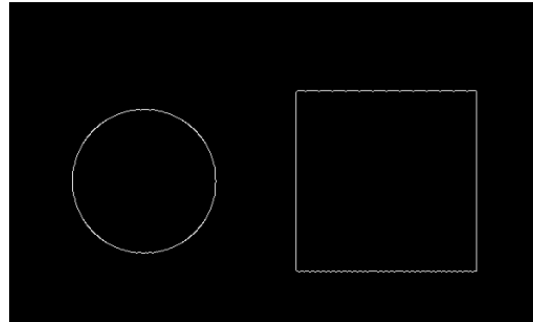
3. CLAHE Local Normalization



4. Direct Canny (Shadow Fragmentation)



5. Global HE + Canny (Noise Artifacts)



6. CLAHE + Canny (Robust Contours)

